

Broken Keyboard

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

While trying to debug his code at 3 am, Bob accidentally knocked over his traditional glass of hot çay right onto his laptop's keyboard.

He managed to dry it off, but his keyboard doesn't work properly anymore. It types lowercase Latin letters normally, but now it has three malfunctioning special keys:

- Backspace now prints `\`. It deletes the last character that was printed on the screen.
- Caps Lock now prints `#`. It toggles the capitalization state on and off. By default, Caps Lock is OFF.
- Shift now prints `*`. It inverts the capitalization of exactly the next keystroke. If a letter was going to be lowercase, it becomes uppercase, and vice versa.

Note: The Shift key state is consumed by the very next keystroke, even if that keystroke is another special key (like Backspace or Caps Lock).

Given a string s representing the exact sequence of keystrokes Bob made, output the correct string that should normally appear if his keyboard wasn't broken. Note that the special keys (`\`, `*`, `#`) are not printed on the screen themselves.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The only line of each test case contains a string s ($1 \leq |s| \leq 100$) consisting of lowercase Latin letters and the characters `\`, `#`, and `*`.

Output

For each test case, output the string that should normally appear on screen. It is possible that Bob deleted all his characters, in which case you should print an empty line.

Example

standard input	standard output
7	maes
mad\es	HELLOworld
#hello#world	SIX
*s*i*x	SEVEN
#seven#	mADEsIsTheBest
#*made#s*s*is#t#he*best	
ab\\c\	ProgrammingIsFun
*programming*is*boring*fun	